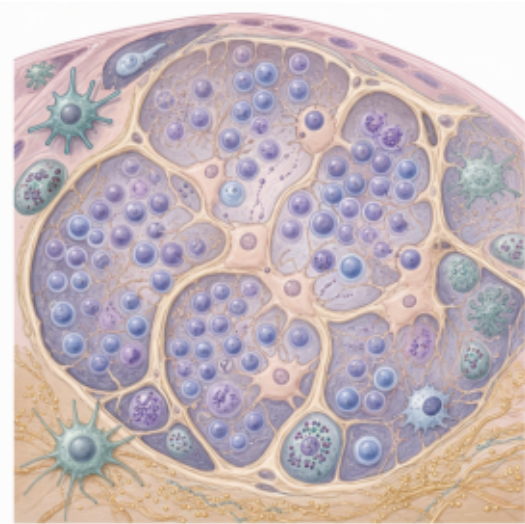


From established organ phenotypes to robustness-filtered pathway hypotheses

Thymus



Established phenotype [6-11]

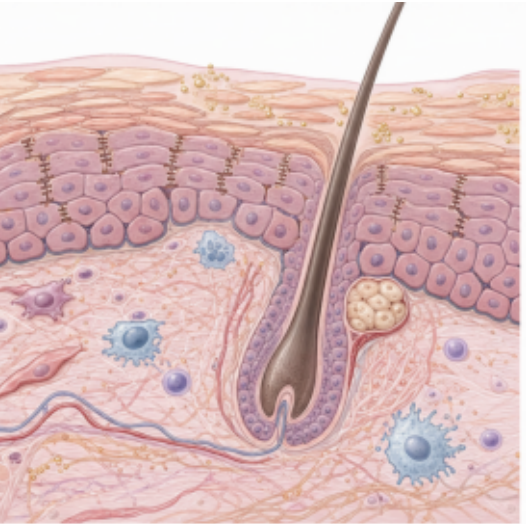
Atrophy, fewer cycling thymocytes, and lower adaptive immune activity.

Robustness-filtered expiMap perspective

Lower repair and cytoskeletal programs are triangulated; lower lymphoid-stromal interaction is an internally robust model-specific signal.

Repair and cytoskeletal state lower; niche interaction lower

Skin



Established phenotype [12-16]

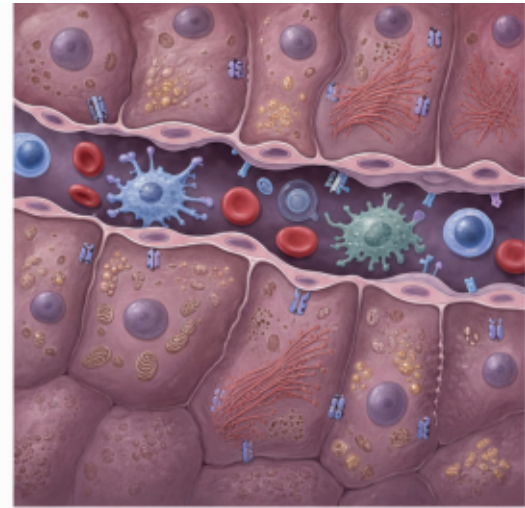
Barrier injury, inflammation, oxidative stress, and impaired wound repair.

Robustness-filtered expiMap perspective

Chromatin, repair, Hedgehog, and barrier-lipid programs are reproducibly lower; cell-junction organization is an added model-specific layer.

Regulatory, regenerative, barrier-lipid, and junction state lower

Liver



Established phenotype [17-23]

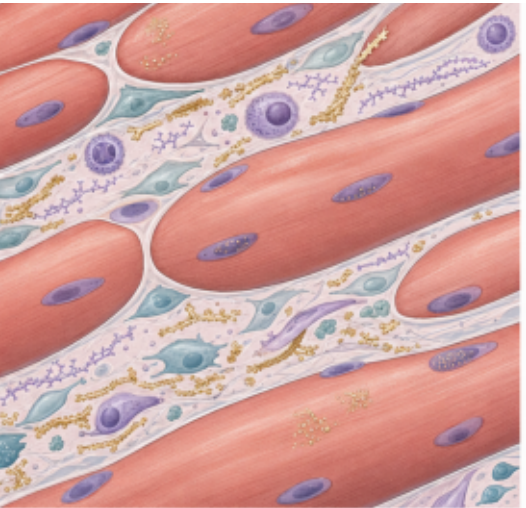
Lipid, xenobiotic, insulin-related, and mitochondrial dysregulation.

Robustness-filtered expiMap perspective

MHC II antigen-presentation and T-cell receptor programs are reproducibly lower; cytoskeletal and matrix directions remain model-sensitive.

Adaptive-immune communication lower

Soleus



Established phenotype [24-30]

Atrophy, fiber-type transition, mitochondrial stress, and contractile change.

Robustness-filtered expiMap perspective

No reviewed pathway passes all conventional, held-out, seed, and composition checks; matrix programs remain follow-up candidates.

No robust complementary pathway direction